Ryan Qi

ryan.qi@mail.utoronto.ca linkedin.com/in/ryanqi github.com/ryanq04

Dual U.S.-Canadian citizen seeking a full-time role

Education

University of Toronto **September 2022 – May 2027**

Bachelor of Applied Science in Electrical Engineering *St. George Campus*

- GPA: 3.82/4.00

Technical Skills

Design & Simulation: PCB Design, Altium Designer, SPICE, Keysight ADS, MATLAB
Testing & Validation: Signal Integrity, Post-Silicon Validation, VNA, TDR, Oscilloscopes, Lab Automation
Programming & Development: C, Python, Verilog, CI/CD
Technical Areas: Power Electronics, Microwave Electronics, Embedded Systems, Control Theory

Experience

Post-Silicon Validation Engineer **May 2025 – August 2026**

Qualcomm – Low Power AI and Audio *Markham, ON*

- Scaled power acquisition across low power validation teams in Canada, U.S., and India; standardized test setups, coverage, and tracking across chip variants to detect inconsistency across FPGA emulations, simulations, and post-silicon. Recognized for finding an NPU hardware power-saving opportunity that directly resulted in a **critical engineering change order**.
- Independently designed and deployed an automated power validation framework using power test firmware, a Python backend and controller, remote testbench automation, a 500 GB database, Docker, and a user-facing frontend. Automated data collection, analysis, storage, and visualization; reduced a 30-min per test manual process to 3 min per test unattended across an expanding coverage of 30+ test vectors, 50+ SoCs, and 5+ chip variants.
- Developed HALs, drivers, and test code in C for memory-constrained systems. Utilized JTAG debugger, scatter file configuration, low observability conditions, and assembly-level debug.
- Oversaw the validation of NPU, power domain sleep/wake/low-power state transitions and operations, resource retention in low-power modes, NoC bandwidth and latency, PLL stability, jitter/phase noise, and AI model performance.

Product Development Engineer **May 2024 – August 2024**

Ford Motor Company – Automotive Networking *Dearborn, MI*

- Characterized 1000BASE-T Automotive Ethernet and High Speed CAN networks through SPICE simulations and lab testing; analyzed eye diagrams using high-speed oscilloscopes, S-parameters using vector network analyzers (VNAs), and Time Domain Reflectometer impedance measurements.
- Verified OEM physical layer and data link layer for component approval, managing 100+ hardware and firmware automotive-grade requirements (e.g., component specifications, electromagnetic compatibility, and ISO standards).
- Analyzed and documented competitor tear-downs of vehicle harnesses and electronic units to identify design choices for cost-saving and quality improvement opportunities.

Electrical Systems Team Member **August 2023 – Present**

aUToronto Design Team – Electrical Systems *Toronto, ON*

- Designed PCB and firmware for power distribution controller (12V, 40A across 20 channels). Used a Teensy 4.1 microcontroller for switch control, power monitoring, and CAN bus communication. This hardware and firmware supported the team's consecutive victories since 2024, and is currently used for autonomous driving on public roads.
- Conducted circuit simulations, soldered, and tested circuits using oscilloscopes, multimeters, and power supplies; developed state machine code and test sequences for validation and stress testing.

Research Assistant **June 2023 – September 2023**

Tsinghua University, Department of Electronic Engineering *Beijing, China*

- Supported design and data acquisition of EEG-based driver mental workload study for Advanced Driver Assistance Systems. Performed literature reviews, created datasets, processed neural signals, and tested AI classification models.

Projects

Automated PCB Tester **March 2025 – April 2025**

- Designed an automated circuit testing device using an STM32 Arm Cortex-M4 platform to acquire and analyze signals. Integrated motor control, a camera interface, signal processing, and UART communication using C and Python to probe pads with 5 mm accuracy at a sampling rate of 1 MS/s.